

MATH 261A 8/28

Rank. $\text{Top, Man, sh } *$
 Groupoid should have finite obj + finite products $\Rightarrow$ can define group object

Def. Group object in $\mathcal{C}$ is (G, m, e, i) where $G \in \text{Ob } \mathcal{C}$,
 $m: G \times G \rightarrow G$, $e: * \rightarrow G$, $i: G \rightarrow G$ s.t. axioms encoded
 e.g. identity = $G = G \times * \xrightarrow{\text{id}_G \times e} G \times G \xrightarrow{m} G$
 compatible atlases forms an equiv. relation

Def. G top. grp if G group obj. in Top .
 in field $\mathbb{R}$ } complete atlas
 then reg. types } an of flow

8/30 Def. A atlas A on M is of regularity type $\mathbb{C}^k$ or real/complex analytic if all transition maps

Rank. We put a structure (an equiv. class of regular atlases on M) on M - not a property.

Ref. $f: X \rightarrow Y$ homeomorphism is diffeomorphism if it preserves regularity of transition functions

Def. $f: X \rightarrow K$ is regular function if $f \circ \psi: \psi(U \cap X) \rightarrow K$ is reg. for any chart.

Rank. $\mathcal{O}_X(U) =$ algebra over K of regular functions on U under $\mathcal{O}_X: (\text{Open } X)^{\text{op}} \rightarrow \text{Ab}$
 X is reg. $f \circ g \Rightarrow \exists W \subset U \cap V$ s.t. $f|_W = g|_W$
 $f, g \in \mathcal{O}_X(V)$

Def. A germ of a regular function $f \in \mathcal{O}_X(U)$ @ $p \in U$ is an equiv. class of f under $\sim$

Rank. $\mathcal{O}_{X,p} =$ algebra of germs of regular funs @ p , i.e. $\mathcal{O}_{X,p} = \lim_{p \in U} \mathcal{O}_X(U)$.

Def. A derivation on $\mathcal{O}_{X,p}$ is a linear map $D: \mathcal{O}_{X,p} \rightarrow K$ s.t. $D(fg) = f(p)D(g) + D(f)g(p)$.

Def. $T_p X$ tangent space = v.s.p. of all derivations @ p .
 Really, there are germs.

Lemma. $(x_i) @ p \Rightarrow T_p X = \text{span}\{D_i\}$ for $D_i(f(\vec{x})) = \frac{\partial}{\partial x_i} f(0)$. ($f \in \mathcal{O}_{X,p} \Rightarrow f \in \mathcal{O}_X(U)$ s.t. $p \in U$)

Pr. $D(x_i) = e_i \in K$. Let $D' = D - \sum e_i D_i$. wts $D' = 0$ so D spans. } view f as mult chart around (x_i)

Further For this, note if x_i small w/ t , $f(x_1, \dots, x_n) = f(0) + \int_0^1 \frac{d}{dt} f(tx_1, \dots, tx_n) dt$ and $\frac{d}{dt} f(tx_1, \dots, tx_n) = \sum x_i \frac{\partial}{\partial x_i} f(x_1, \dots, x_n, t)$ w/ initial $h_i(x_1, \dots, x_n) = 0$
 $D'(f) = \sum (D'(x_i) h_i(0) + x_i(0) D(h_i)) = \sum D'(x_i) \cdot h_i = \sum (D(x_i) - \sum e_j D_j(x_i)) h_i$

9/4 Def. $f: X \rightarrow Y$ is reg. if $\forall U \subseteq Y$ and $h \in \mathcal{O}_Y(U)$, $V = f^{-1}(U) \subseteq X$ so $h \circ f \in \mathcal{O}_X(V)$.

Rank. regular maps are maps that are regular (from last time) is local coord.

Def. $F: X \rightarrow Y$ reg. differential $df_p: T_p X \rightarrow T_{F(p)} Y$ is linear map s.t.
 for any $f \in \mathcal{O}_{Y, F(p)}$ and $v \in T_p X$, $df_p(v)(f) = v(f \circ F)$.

Rk. $(\text{Cof}) = G + Fx$ mod functor $\sim$.

$(0,1) \hookrightarrow \mathbb{R}$ not cl. embedding. $\mathbb{R} \xrightarrow{\Delta} \mathbb{R}$ homeo but not immersion.
 $S^1 \hookrightarrow \mathbb{R}$ cl. embedding. $\mathbb{R} \rightarrow \mathbb{R}^2$ $\nexists$ not homeo.

MATH 261A

Def. $F: X \rightarrow Y$ regular is immersion if $\forall p \in X$, $d_p F$ is injective.
 (consider $F(X)$ submanifold in Y) (injection in particular).
 Def. F embedding if immersion and $F: X \rightarrow F(X)$ homeomorphism.

Def. F cl. embedding if $F(X) \subset Y \Rightarrow F(X)$ cl. submanifold. (e is not reg. map).
 i.e. m, n are regular maps.
 Def. A C^k , real, or complex Lie group is a group eqn in set of maps of corresponding type.
 Hilbert's Fifth Problem: All Lie groups are analytic.

Def. Lie group morphisms are regular group homomorphisms (pseudo-).
 $x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_n^2 = 0$

Def. $O(p, q)$ is classical groups, orthogonal groups $O(p, q)$ are linear maps $\mathbb{R}^{p+q} \rightarrow \mathbb{R}^{p+q}$ such that $A^T = -A$.
 Def. symplectic group $Sp_{2n}(\mathbb{R}) = \{A \in GL_{2n}(\mathbb{R}) \mid A^T J A = J\}$ for $J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$.
 $\{AA^T = I\}$.

Rank $U(n)$ is a real Lie group. The conjugate is not held.

Rank. $G \xrightarrow{\phi} H$ Lie gp morph $\Rightarrow L_{\phi} \circ \phi_* = \phi_*(L_g \cdot)$ $\xrightarrow{\text{Lys iso}}$
 $\Rightarrow (d\phi_g) \circ \phi_* = \phi_* \circ d\phi_g \Rightarrow L_g \text{ diff iso } \phi_* \text{ same rank as } (d\phi_g)_g$.
 $\Rightarrow$ all Lie group morphisms are constant rank.

Rank Bijections Lie group morphisms $\Rightarrow$ diffeomorphism.
 Pf. $G \times H$ Lie $\Rightarrow$ constant $\Rightarrow \phi(x_1, \dots, x_n) = (x_1, \dots, x_n, 0, \dots, 0)$, ϕ is so $n = m$ and clearly smoothly invertible locally so ϕ has regular inverse, so ϕ diffeo.

Def. Closed Lie subgroup is $H \subset G$ that's embedded submanifold.
 Def. Closed Lie subgroup $\Leftrightarrow$ closed subgroup ($H = \overline{H}$ subgroup).

Def. Representation of G is a sp. V s.t. $G \curvearrowright V$ is regular. $J \in \mathfrak{g} \mapsto A(g)$
 Homomorphisms (intertwining operators) is $A: V \rightarrow W$ s.t. $A \circ \rho_g = \rho_h \circ A$.

Def. Rep G = cat. of finite dim reps of G , + duals + tensor products.

$$V \otimes^{G \times V} W \cong \mathfrak{g}_W \cdot V \subset V$$

$$(g \cdot v^*)(w) \\ v^*(w \cdot g^{-1})$$

MATH 261A

9/9 th.

$$G \curvearrowright V \leadsto G \curvearrowright V^* \text{ by } g_{V^*} = (g_V^{-1})^* \\ g_{V \otimes W}(v \otimes w) = g_V \otimes g_W \\ g_w(w)$$

Rank. In $\text{Rep}_R G$, we want $\text{duals} + \text{tensors} + \text{subrep} + \text{quot}$. $\rightarrow g_w(w)$

Thm. $G \curvearrowright X, x \in X$. 1) $G \cdot x \subset X$ invariant subset 2) G_x cl. Lie subg. of G 3) G_x 4) $G \cdot x$ is orbit $\Rightarrow G/G_x \rightarrow G \cdot x$ is diffeo. (if G finite and x is G -reg.)

9/11 Ex. $\mathbb{R} \curvearrowright T^2 = \mathbb{R}^2/\mathbb{Z}^2$ by $t \cdot (x, y) = (x+t, y+t\alpha)$. $\alpha \notin \mathbb{Q} \Rightarrow \mathbb{R} \cdot (x, y)$ is not embedded

Cov. (to handle lifting prop.). $\mathbb{Z}$ simp. conn. $\Rightarrow \exists \tilde{f}: \tilde{X} \rightarrow X$ lift $\tilde{f}$ by path lifting. $\tilde{f}(z) = \tilde{b}$

9/13 Rank $\pi_1(X, b) \curvearrowright p^{-1}(b) \cong \pi_1(\tilde{X}, \tilde{b})$ get unique lift $\tilde{\alpha}$ of α s.t. $\tilde{\alpha}(0) = \tilde{b}$. if Y path action is transitive. if $\pi_1(Y) = 1$, then action is free. $\tilde{b}, \tilde{b}' \in Y$. $\alpha: \tilde{b} \rightarrow \tilde{b}'$. $[p \circ \alpha] \cdot \tilde{b}$ lifts to $\tilde{b}' \Rightarrow p^{-1}(b) \xrightarrow{b \mapsto \tilde{b}} \pi_1(X, b)$

Rank. $S^1 \rightarrow \mathbb{R}P^1, \mathbb{R} \rightarrow S^1$ universal covers w/ kernel $\mathbb{Z}_2, \mathbb{Z}$ respectively. lift $\tilde{\alpha}: \tilde{S}^1 \rightarrow \tilde{S}^1$

Thm. $\tilde{G} \xrightarrow{p} G$ universal cover has Lie group structure. $\tilde{G} \times \tilde{G} \rightarrow G \times G \rightarrow G$ lift $(\tilde{g}, \tilde{h}) \rightarrow \tilde{gh}$. Pf. pick $\tilde{1} \in p^{-1}(1) \subset \tilde{G}$. now, $\tilde{G} \times \tilde{G} \rightarrow G \times G \rightarrow G$ lift $(\tilde{1}, \tilde{1}) \rightarrow \tilde{1}$. also get p is homeo via $p(\tilde{m}(\tilde{g}, \tilde{h})) = m(p(\tilde{g}), p(\tilde{h}))$

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin} \rightarrow \text{SO} \rightarrow 1$$

Def. $\text{Spin}(n)$ is defined as universal cover of $\text{SO}(n)$ p is 2-sheet covering s.t. $\pi_1(\text{SO}(n)) \cong \dots \cong \pi_1(\text{SO}(3)) = \mathbb{R}/\mathbb{Z} = \mathbb{Z}_2$

Def. Lie Algebr of G is $\mathfrak{g} = T_1 G = \text{Lie}(G) = \{ \text{left-invariant v.f. on } G \}$

9/16 Rank All closed groups are defined by $\det A = 1$, and A is unimodular like for B_n

9/20 Rank. $\text{Ad}_g: G \rightarrow G \leadsto \text{Ad}_g: \mathfrak{g} \rightarrow \mathfrak{g}$. let $\text{ad}_x: \mathfrak{g} \rightarrow \mathfrak{g}$ $\text{ad} = \text{Ad}_*$ Prop. $\text{Wad}: \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$, $\text{Ad}: G \rightarrow \text{GL}(G)$, $\text{ad} = \text{Ad}_*$ (2) $\exp(\text{ad}) = \text{Ad}(\exp)$ Pf.

MATH 261A

9/25 Prop. Given $G \curvearrowright X$, we get $\text{act}: \mathfrak{g} \rightarrow \text{Vec}(X): \xi \mapsto \frac{d}{dt}|_{t=0} (f(\exp(-t\xi) \cdot x))$
 cl. Lie algy $(x \in X, \mathfrak{g}_x \text{ is stabilizer})$

Thm. (Orbit-stabilizer). 1) $\mathfrak{g}_x \leq \mathfrak{g} \Rightarrow \text{Lie}(\mathfrak{g}_x) = \mathfrak{g}_x = \{\xi: \eta_\xi(x) = 0\}$. $\mathfrak{g} = \mathfrak{g}_x \oplus \mathfrak{u}$,
 2) $\mathfrak{g}/\mathfrak{g}_x \xrightarrow{\varphi} T_x X$, get $\varphi: \bar{\xi} \mapsto \xi \cdot x$ is an isomorphism. $\mathfrak{u} \hookrightarrow T_x X$
 if φ is embedding then $\mathfrak{g}/\mathfrak{g}_x \cong \mathfrak{g} \cdot x$. $\mathfrak{u} \xrightarrow{\varphi} T_x X$
 $\mathfrak{u} \xrightarrow{\varphi} T_x X$
 $\mathfrak{u} \xrightarrow{\varphi} T_x X$

Cor. (Isomorphism). $G \curvearrowright X$ L.G. non $\Rightarrow \mathfrak{g}_x \leq \mathfrak{g}$ cl. Lie algy, $\bar{\varphi}: \mathfrak{g}/\mathfrak{g}_x \rightarrow T_x X$ isomorphism.

Prop. $\mathfrak{g} \rightarrow \mathfrak{g}_x$ accounts Lie algebra to Lie algebra $\text{Lie}(\mathfrak{g}_x) = \mathfrak{g}_x$.

Ex. (i) take $V \in \mathbb{R}^n$ f.d. $G_v = \{g: g \cdot v = v\}$. $\text{Lie}(G_v) = \mathfrak{g}_v = \{\xi \in \mathfrak{g}: \xi \cdot v = 0\}$.

(ii) A f.d. algebra $\mathfrak{a}/\mu: \mathfrak{a} \otimes \mathfrak{a} \rightarrow \mathfrak{a}$, $\mathfrak{a} \otimes \mathfrak{a} \cong \mathfrak{gl}(\mathfrak{a})$, if $\mathfrak{a}$ cl. Lie algebra $\mathfrak{a} \otimes \mathfrak{a} = \mathfrak{gl}(\mathfrak{a})$
 $\text{Lie}(\mathfrak{g})$ $\mathfrak{a} \otimes \mathfrak{a} \rightarrow \mathfrak{gl}(\mathfrak{a})$ $d(\xi) = \xi \cdot d(\eta) + (d\xi) \cdot \eta$

Prop. $\text{Lie}(\text{Ad}(A)) = \text{Der}(\mathfrak{a}) \subset \mathfrak{gl}(\mathfrak{a})$. $\mathfrak{a} \in \mathfrak{a} \otimes (\mathfrak{a} \otimes \mathfrak{a})^*$
 Pf.

Def. $Z(\mathfrak{g}) = \{\xi \in \mathfrak{g}: \xi \eta = \eta \xi \text{ } \forall \eta \in \mathfrak{g}\}$, $\mathfrak{g}(\mathfrak{g}) = \{\xi \in \mathfrak{g}: \forall \eta \in \mathfrak{g}, [\xi, \eta] = 0\}$.

Prop. $Z(\mathfrak{g}) \subset \mathfrak{g}$ cl. Lie algy $\Rightarrow \text{Lie}(Z(\mathfrak{g})) = \mathfrak{g}(\mathfrak{g})$.

Pf. $\xi \in Z(\mathfrak{g}) \Rightarrow \exp(\xi) \exp(\eta) = \exp(\eta) \exp(\xi) \text{ } \forall \eta \in \mathfrak{g} \Leftrightarrow \exp(\text{ad}_\xi \eta) = \exp(\eta) \Leftrightarrow \xi \in \mathfrak{g}(\mathfrak{g})$
 $\xi \in \mathfrak{g}(\mathfrak{g})$

Def. $\mathfrak{g}/Z(\mathfrak{g})$ is the adjoint group of $\mathfrak{g}$.

Thm. (Fundamental Thm of Lie Theory). (1). Lie group G . $\exists$ bij. Lie subgroups & subalgebras $(\mathfrak{h} \leq \mathfrak{g}, \mathfrak{g} \leq \mathfrak{g})$.

(2). G simply connected $\Rightarrow \text{Hom}(G, M) \xrightarrow{\text{bij.}} \text{Hom}(\mathfrak{g}, \mathfrak{m})$.

(3). Any f.d. Lie algebra is the Lie algebra of some Lie group $\Rightarrow$ of some simply conn. Lie group.
 Cor. Integer-valued exponential Lie group is f.d. Lie algebra. $\mathfrak{g} \in \mathfrak{g}$

Def. k -dim distribution on X is k -dim subbundle of TX s.t. locally, on $U \subset X$, $D = \text{span}(V_1, \dots, V_k)$

Def. D integrable if $\forall x \in X, \exists$ neighborhood $U \subset X$ and local coords $x_1, \dots, x_n$ on U s.t. $D = \text{span}(\partial_{x_1}, \dots, \partial_{x_k})$

Prop. D int. $\Leftrightarrow \forall x \in X, \exists$ integral submanifolds $S_x \subset X$ (immersed).
 called a leaf s.t. $T_y S_x = D_y \text{ } \forall y \in S_x$. $\rightarrow = \{y \in X: \exists \gamma: [0, 1] \rightarrow X \text{ s.t. } \gamma(0) = x, \gamma'(t) \in D_{\gamma(t)}\}$

Def. integrable distribution on foliations. $X = \bigcup_x S_x$. $\mathfrak{g}(\mathfrak{g})$

f. two on the

Ex. (i) orbits of $R \curvearrowright T$ on $\mathbb{R} \times \mathbb{R} \rightarrow (x, y) \mapsto (x, y + \alpha x)$ (ii) fiber bundles $\Rightarrow$ leaves of the foliation
(iii) rank 2 dist $\Rightarrow$ direction field $\Rightarrow$ integral solutions = integral curves.

(Nm-) Ex. $\mathbb{R}^3, v = \partial_x, w = x\partial_y + \partial_z \Rightarrow [v, w] = \partial_y \notin \mathcal{L} = \text{sp}(\tilde{w}_2, w) \Rightarrow \mathcal{L}$ not integrable.

$\Rightarrow D_{f \circ g} = \text{sp}(\{f_i : i \in I\}) = \{ \sum f_i z_i \}$. if $g_j = \langle h_j \rangle$, then

$$[\sum f_i z_i, \sum g_j z_j] = \sum_{i,j} (f_i(z_j, 0_j) \circ z_j - g_j(z_i \circ f_i) z_i + f_i z_j [z_i, z_j])$$

$$\Rightarrow D_{f \circ g} = D_{f \circ g} + \sum f_i z_i$$

$D^2 \text{ int } \gamma \cdot D^2 \gamma = D^2 \gamma \Rightarrow$ if $\{S_\gamma\}$ is solution, then $S_\gamma = 0 \cdot S_1$. $S_1 = M$.
 and $\text{cl}_\gamma, T_1 M = T_1 S' = \gamma$. $\square$

$\begin{matrix} & K & \\ p_1 \swarrow & & \searrow p_2 \\ G & & H \end{matrix}$
 Pf. pick $p: g \rightarrow h$, def $\sigma_{g,h} = 1 \times p \circ \sigma_g = g \circ p$
 let $K \leq G \times H$. $\text{Lie}(K) = \text{im } \sigma$.
 $\begin{matrix} \text{crossing} \\ \text{simplex} \end{matrix} \quad \{4u(g, h)\}$

$(p_1)_* \circ \sigma = \text{id}_Y$ and $\sigma \circ (p_1)_* = \text{id}_X$ $\Rightarrow (p_1)_*$ bij so conway $\left. \begin{array}{l} \text{line 1} \\ \text{line 2} \end{array} \right\} \sigma(q \neq p)$

but G ring. case $\Rightarrow K = G$ *24/11/2023*

so $\begin{matrix} K & \xrightarrow{p_i} & K \\ \downarrow \sigma & & \downarrow \psi \end{matrix} \rightarrow \ell_* = \psi$ as we wish. $\square$

Thm. (Ado). any f.d. Lie algebra $\mathfrak{g}$ is Lie isomorphic to $\mathfrak{gl}(V)$ for some f.d. V
(i.e. $\exists$ faithful rep of $\mathfrak{g}$)

✓ Prop. if $z(g) = 0$, the ad: $g \rightarrow \mathfrak{gl}(g)$ is inj and sur. $\downarrow$ repr.

$\forall$ f.d. $\mathfrak{g}$, $\exists V$ v.s.p. s.t. $G \cong \text{Lie}(V)$ s.t. $\mathfrak{g} = \text{Lie}(G)$. $\rightarrow$ if we take $G \cong \tilde{G}$, then apply

Def. $\gamma_C = \gamma \otimes_C C$ is corestriction of γ . γ is restriction of γ_C to $\mathcal{A}$.

so $K_{\mathbb{C}} \cong \mathbb{C}$ is a $\mathbb{C}$ -algebra. multiplication in $\text{gl}_n(\mathbb{R})_{\mathbb{C}} = \text{gl}_n(\mathbb{C}) = \text{Mat}_n(\mathbb{C})$ is the same as for $\text{gl}_n(\mathbb{R})$ but with $\mathbb{C}$ instead of $\mathbb{R}$.

MATH 261A.

ALIAS NTAM

9/27 th.

Def. G con. complex Lie group, $K \subseteq G$ real Lie group s.t. $\text{Lie}(K) \otimes_{\mathbb{R}} \mathbb{C} \cong \text{Lie}(G)$, then K is a real form of G .

Def. K simp. con. Lie group. $\mathfrak{g} = \text{Lie}(K) \otimes_{\mathbb{R}} \mathbb{C}$. $\exists$ a simple con. Lie group G w/ $\text{Lie}(G) = \mathfrak{g}$. G is complexification of K .

Proof. $\text{SL}_2 \mathbb{R} \xrightarrow{\text{2-fold}} \text{SL}_2 \mathbb{C}$ always used initially but not always inj.
 $\text{SL}_2 \mathbb{C} \hookrightarrow \text{SL}_2 \mathbb{C} \supset \text{SL}_2 \mathbb{R} \otimes \mathbb{C}$

9/30 group!

10/2

Rank. $V \otimes V = S^2 V \oplus \Lambda^2 V$, $V \otimes V^* = S^2 V^* \oplus \Lambda^2 V^*$.

Rank. $\text{Rep}_K \mathfrak{g}$, $\text{Rep}_K G$ are abelian categories closed under the operation $\Rightarrow$ still rep.

Lemma (Schur). V, W irr. $\Rightarrow \text{Hom}_{G \text{ or } \mathfrak{g}}(V, W) = \begin{cases} 0 & V \not\cong W \\ 1 & V \cong W \end{cases}$ general version is: for dim K , K alg. cl. when "1" is replaced by $\dim$ Hom is $\dim$ of $\text{End}(V)$
 Pf. say $\phi \in \text{Hom}_G(V, W)$. Ker ϕ subrep of V irr, ker $\phi = 0$ on V . if $0 \Rightarrow$ inj (same arg for irr ϕ no also surj).
 how, $\phi: V \rightarrow W$ has signature w/ $\text{ker } \lambda$ in $\phi - \lambda I$ but inj so V to W is $\phi - \lambda I$, so $\phi = \lambda I$, so ϕ is λ times id .

Cor. V irr $\Rightarrow Z(G \text{ or } \mathfrak{g}) = \{ \lambda \text{id} : \lambda \in \mathbb{C} \} \cap G \text{ or } \mathfrak{g}$.

Cor. irreps of abelian groups/Lie algebras are 1-dim. bc weights can't act by a constant on an irr.

Rank. for completely reducible rep $V = \bigoplus V_i^{m_i}$, $W = \bigoplus W_i^{n_i}$, $\text{Hom}_G(V, W) = \bigoplus M_{n_i \times m_i}$.

Def. $V \in \text{Rep}_G$ is unitary if admits a G -invariant Hermitian inner product. $(x, y) = (gx, gy)$ ($g \in G$).
 invariant $\rightarrow$ $(x, y) = (gx, gy)$ ($g \in G$).
 unit form $\rightarrow$ $(x, y) = (gx, gy)$ ($g \in G$).
 $\phi = (x, y) + (y, x) = 0$
 $Z \in \mathfrak{g}$.

Prop. any unitary rep is completely reducible.

Pf. V unit. rep. $W \subseteq V$ subrep. we claim $W^\perp$ is a subrep, i.e. $u \in W^\perp, w \in W \Rightarrow (gu, w) = (u, g^{-1}w) = 0$. $\hookrightarrow g \cdot u \in W^\perp$.
 using ϕ subrep on $V = W \oplus W^\perp$.

10/2 etc.

MATH 261A

group (16/5a).

Prop. G fin. $\Rightarrow \forall e \in \text{Rep } G$ unitary $\Rightarrow$ completely reducible

PF. start w/ any non-deg Hermitian form B . Let $\tilde{B}(v, w) = \frac{1}{|G|} \sum_g B(gv, gw)$.
 $\tilde{B}$ still non-deg, inv., Hermitian form. on G rep unitary
~~(G-invariant)~~ ^{inv.}

Prop. Unitary structure on V unique up to mult. by positive constant.

PF. B_1, B_2 unitary struct we get $B_1 = A^t B_2$ for $A: V \rightarrow V$ on $B_1(v, w) = B_2(Av, w)$.
 we'll show A is identity so get unit (by Schur's lemma). $\Rightarrow A = \lambda \text{Id}$, $B_2(v, w) \Rightarrow \lambda \in \mathbb{R}_{>0}$

$$B_2(Agv, w) = B_1(gv, w) = B_1(v, g^{-1}w) = B_2(Av, g^{-1}w) = B_2(gAv, w)$$

$$\text{right } G\text{-invariant} \Rightarrow \int_G f(g) dg = \int_G f(g) dg$$

Prop. we get Haar measure dg o.t. on cpt Lie groups G as $\int_G 1 dg = 1$

Prop. we can adapt the above as $\tilde{B}(v, w) = \int_G B(gv, gw) dg$.

Prop. $SL_n \subset SO_n$ and $SO_n \subset SL_n$ - how can we see this as a consequence of $SO(n)$.

Hermitian form...

10/16. $\text{rad}(\mathfrak{g}) =$ maximal solvable ideal of $\mathfrak{g}$, $0 \rightarrow \text{rad}(\mathfrak{g}) \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/\text{rad}(\mathfrak{g}) \rightarrow 0$
 $\text{rank } \mathfrak{g}$ is semisimple (s/s) if $\text{rad}(\mathfrak{g}) = 0$. (split in char. 0).

Prop. in char $k=0$, $k = k$ (e.g. $k = \mathbb{C}$). $V \in \text{In Rep } \mathfrak{g} \Rightarrow \text{rad}(\mathfrak{g})$ acts on V by

PF. $\exists v \in V$, $\lambda: \text{rad}(\mathfrak{g}) \rightarrow k$ s.t. $\forall a \in \text{rad}(\mathfrak{g})$, $av = \lambda(a)v$. (Schur's lemma)

alg. cl. pick any $x \in \mathfrak{g}$, $W = \langle v, xv, x^2v, \dots \rangle$. $\mathfrak{g}_x = \text{rad}(\mathfrak{g}) + \langle x \rangle$ acts on W .
 (if x is nilpotent). $\forall a \in \text{rad}(\mathfrak{g}) \Rightarrow ax^i v = \lambda(a)x^i v + \sum_{j=0}^{i-1} \binom{i}{j} (ax)^j v$.
 $\forall y \in [\mathfrak{g}, \text{rad}(\mathfrak{g})]$, $0 = \text{tr}(y|_W) = \lambda(y) \dim W \Rightarrow \text{tr}(y|_W) = 0 \Rightarrow \lambda(y) = 0$.
 $\Rightarrow ax^i v = \lambda(a)x^i v \quad \forall a \in \text{rad}(\mathfrak{g}) \Rightarrow \forall x = \text{ad } y$ for $\text{rad}(\mathfrak{g})$ is $\mathfrak{g}$ in sub. of $V_\lambda = V$.

Df. $\mathfrak{g}$ is reductive if $\text{rad}(\mathfrak{g}) = \mathfrak{z}(\mathfrak{g})$, i.e. $[\mathfrak{g}, \text{rad}(\mathfrak{g})] = 0$.

Prop. (Levi decomposition) over $\mathbb{C} \Rightarrow$ reductive = semisimple $\oplus$ abelian.

Df. $\mathfrak{g}$ is invariant under ρ if $B([x, y], z) + B(y, [x, z]) = 0$.

Ex. $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, $B_V(x, y) = \text{tr}(\rho(x)\rho(y))$

10/16 std

Prop. If B is invertible, then $I \subset \mathfrak{o}_F$ ideal, then $I^\perp \subset \mathfrak{o}_F$ ideal.

Pf. Use orthogonal basis is also orthonormal, and I is a sublattice under adjoint rep. $\text{Rep } \mathfrak{o}_F$

Prop. B_V nondegenerate for some rep $V \Rightarrow \mathfrak{o}_F$ is reduction.

Pf. wts $[\mathfrak{o}_F, \text{rad}(\mathfrak{o}_F)] = 0$. Let us know $[\mathfrak{o}_F, \text{rad}(\mathfrak{o}_F)]$ acts by 0

on irr. representations. get composition factors $V_1, \dots, V_k$ of $V \rightarrow$ then $B_V(x, y) = \sum_{i=1}^k B_{V_i}(x, y)$. Let $x \in [\mathfrak{o}_F, \text{rad}(\mathfrak{o}_F)]$
 $\Rightarrow x|_{V_i} = 0 \Rightarrow B_{V_i}(x, y) = 0 \forall y \in \mathfrak{o}_F \Rightarrow x \in \ker B_V \Rightarrow x = 0$.
 $\Rightarrow B_V$ nondegen

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

Ex. $V = k^n$, $\text{gl}_n(k) = \text{gl}_n(V)$, $B_V(E_{ij}, E_{kl}) = \delta_{jk} \delta_{il}$.

~~B_V nondegen $\Rightarrow \text{gl}_n(k)$ reduction. Let, can~~
 already computed $\chi(\text{gl}_n(k)) = k^n$ scalar matrix $\Rightarrow$ justifying
 this is only filtration of comp factors but subspaces $\Rightarrow$ $\text{gl}_n(k)$ semisimple.

1	0	0
0	1	0
0	0	1

Prop. All derivations in Algebras are reduction

Pf. $\mathfrak{o}_F \subset \text{gl}(V)$, consider natural matrix rep V , B_V nondegen (how to check over $\mathbb{C}$ and $\mathbb{R}$).

Def. Killing form $K_{\mathfrak{g}} = K$ on $\mathfrak{g}$ is $K(x, y) = \text{tr}(\text{ad } x, \text{ad } y)$.

$\Rightarrow \text{char } K = 0$, $\mathfrak{g}$ semisimple $\Rightarrow \text{rad} = 0$.

Thm. (Cartan's Thm of solvability). $\mathfrak{g}$ solvable iff $[\mathfrak{g}, \mathfrak{g}] \subset \text{ker}(K)$.

Thm. (Cartan's Thm of semisimplicity). $\mathfrak{g}$ is semisimple iff K is nondegenerate.

Jordan Decom.

Prop. $A \in \text{gl}_n(V)$, $\text{char}(K) = 0$ can be written uniquely as $A = A_s + A_n$.
 where $A_s, A_n = A_n$. A_s has $\exists p(x) \in K[x]$ s.t. $A_s = p(A)$. $\Rightarrow$ nilpotent.
 $K[x]/(f) \rightarrow \text{char poly} \rightarrow$ break up vs.

Pf. $V_K \in K[x]$ -modules, $V_K \cong K[x]/(x-\lambda_1)^{n_1} \oplus \dots \oplus K[x]/(x-\lambda_m)^{n_m}$ from Chinese remainder thm + decomposition PID.
 $\exists p(x) \in K[x]$ s.t. $p \mapsto \lambda_i \pmod{(x-\lambda_i)^{n_i}}$
 $\Rightarrow p(A) = p\left(\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}\right) = \begin{bmatrix} p(\lambda_1) & 0 & 0 \\ 0 & p(\lambda_2) & 0 \\ 0 & 0 & p(\lambda_3) \end{bmatrix} \rightarrow$ makes it diag, with $\lambda_i = A_s$.

λ_1	0	0
0	λ_2	0
0	0	λ_3

clear that $A_s = p(A) \Rightarrow A_n = A - p(A)$, $A_p(A) = p(A)A$ in A_n $A_s = A_s A_n$ is.

Action of $\text{GL}(F/K)$ on A_s, A_n preserves uniqueness: $A = A'_s + A'_n \Rightarrow A'_s A = A'_s \Rightarrow A'_s p(A) = p(A) A'_s$.
 A'_s, A'_n but A_n is nilpotent $\Rightarrow A_s, A'_s$ can be diagonalized by the same basis $\Rightarrow A_s - A'_s = A'_n - A_n \in \text{nil}$.
 the same $\Rightarrow$ stable under action $\Rightarrow$ they are the same!! $\Rightarrow$ must be

21
10/10/10

MATH 261A

already
 $k(x, y) = t(\log)$

no more
matched ident.

angle

Def. $\mathfrak{g}$ is simple iff Killing form is nondegenerate iff $\mathfrak{g} = \bigoplus_{i=1}^k \mathfrak{g}_i$.

Cor. any quotient/ideal is also simple.

goc

not too simple

too simple.

Def. $\mathfrak{g}/\mathfrak{h}$ s/o iff $\mathfrak{g}/\mathfrak{h} \subseteq \mathfrak{g}$ s/o. $\mathfrak{g}/\mathfrak{h}$ s/o iff $\mathfrak{g}/\mathfrak{h}$ s/o.

Ex. $\mathfrak{g}/\mathfrak{h}$ s/o $\Rightarrow \mathfrak{g}/\mathfrak{h}$ s/o as $\mathfrak{g}/\mathfrak{h} = \mathfrak{g}_1 \oplus \mathfrak{g}_2, \mathfrak{g}_1, \mathfrak{g}_2/\mathfrak{h}$.

$\mathfrak{g}_1$ and ideal in $\mathfrak{g}$ is $\mathfrak{g}_1 = [\mathfrak{g}_1, \mathfrak{g}_1] = \mathfrak{g}_1$ as $\mathfrak{g}_1$ is $\mathfrak{g}$ s/o.

Thm. $\mathfrak{g}$ s/o $\Rightarrow \text{ad}: \mathfrak{g} \rightarrow \text{Der}(\mathfrak{g})$ is injective.

Pf. ad is injective b/c $\mathfrak{g}(\log) = 0$. $\mathfrak{g} \subseteq \text{Der}(\mathfrak{g})$ ideal in

$\mathfrak{g} \subseteq \text{Der}(\mathfrak{g}), \mathfrak{g} \subseteq \mathfrak{g} \Rightarrow [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{g} = [\mathfrak{g}, \mathfrak{g}] - [\mathfrak{g}, \mathfrak{g}] = [\mathfrak{g}, \mathfrak{g}]$

Prop. extend the Killing form to $\text{Der}(\mathfrak{g})$. $\mathfrak{g} \subseteq \text{Rep}(\text{Der}(\mathfrak{g}))$.

$k(x, y) = \text{tr}(\text{ad}(x)\text{ad}(y))$ is not just ad $\text{Der}(\mathfrak{g})$.

$k/\mathfrak{g}$ Killing form is nondegenerate

$I \subseteq \mathfrak{g}^{\perp}$ w.r.t k in $\text{Der}(\mathfrak{g})$, $I \cap \mathfrak{g} = 0$, $I \oplus \mathfrak{g} = \text{Der}(\mathfrak{g})$, $I \subseteq \text{Der}(\mathfrak{g})$ ideal

Cor. $\text{Aut}(\mathfrak{g}) = G$, $\text{Lie}(G) = \text{Der}(\mathfrak{g})$. if $\mathfrak{g}$ s/o, $\text{Lie}(G) = \mathfrak{g}$.

if completely reducible, will split

Prop. extend rep: $0 \rightarrow \mathfrak{u} \rightarrow \mathfrak{v} \rightarrow \mathfrak{w} \rightarrow 0$; $\mathfrak{v} = \mathfrak{u} \oplus \mathfrak{w}$ as vector space.

as $x \in \mathfrak{g}$ $\Rightarrow \rho(x) = \begin{bmatrix} \rho_{\mathfrak{u}}(x) & 0 \\ 0 & \rho_{\mathfrak{w}}(x) \end{bmatrix}$ $x \cdot (u, w) = (x \cdot u + \text{ad}(x)w, x \cdot w)$

where $\rho: \mathfrak{g} \rightarrow \text{Hom}_k(\mathfrak{w}, \mathfrak{u})$.

we have $[x, y](u, w) = x y(u, w) - y x(u, w) = \rho(x)\rho(y)w - \rho(y)\rho(x)w = [\rho(x), \rho(y)]w$

so we get the relation that: $\rho([x, y]) = [\rho(x), \rho(y)]$.

$\Rightarrow \mathfrak{u} \oplus \mathfrak{w}^{\perp} \cong \text{Hom}_k(\mathfrak{w}, \mathfrak{u})$ of $\mathfrak{g}$ via $x \cdot A = [x, A]$.

1 coadjoint w/coff in $\mathfrak{g}$

Def. $E \in \text{Rep}(\mathfrak{g})$, $\mathfrak{z}'(\mathfrak{g}, E) = \{a: \mathfrak{g} \rightarrow E: a(x) = x \cdot a(y) - y \cdot a(x)\}$.

Ex. $\mathfrak{z}'(\mathfrak{g}, k) = (\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}])^*$, $\mathfrak{z}'(\mathfrak{g}, \mathfrak{g}) = \text{Der}(\mathfrak{g})$.

Prop. $\forall a \in \mathfrak{z}(\mathfrak{g}, \text{Hom}_k(\mathfrak{w}, \mathfrak{u})) \rightarrow 0 \rightarrow \mathfrak{u} \rightarrow \mathfrak{v} \rightarrow \mathfrak{w} \rightarrow 0$

Ex. $\mathfrak{v}_0 = \mathfrak{u} \oplus \mathfrak{w}$.

10/21 std.

$$0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$$

Def. Morphism of extensions:
$$\begin{array}{ccccccc} & & \text{id} \downarrow & & \downarrow f & & \downarrow \text{id} \\ 0 & \rightarrow & U & \rightarrow & V & \rightarrow & W \rightarrow 0 \end{array}$$

Prop. if $0 \rightarrow u \rightarrow v \rightarrow w \rightarrow 0$, $f \circ u = u$, $f \circ v = v$, $\text{gr } V = U \oplus W$, $f = \text{id} : U \oplus W \rightarrow U \oplus W$.

Prop. if f is an iso of extensions, then, $f = \begin{bmatrix} A & B \\ 0 & \text{id} \end{bmatrix}$, $A \in \text{Hom}_K(W, U)$ $\left\{ \begin{array}{l} f(u, w) \\ (u, w) \end{array} \right.$

$$\begin{aligned} x \cdot f(u, w) &= (xu + xAw + b(x)u, xw) \text{ let } \\ f(x(u, w)) &= (xu + a(x)w + Axu, xw) \\ \Rightarrow a(x) - b(x) &= xA - Ax = [x, A] = x \cdot A. \end{aligned}$$

(is $\text{gr } V = U \oplus W$ + id)

$\frac{1}{2}$ -cocycle w/ coeffs in E $\left\{ \begin{array}{l} \text{iff } \exists A \in \text{Hom}_K(W, U) \text{ s.t.} \\ a(x) = x \cdot A = [x, A] \end{array} \right.$

Def. $E \in \text{Rep } \mathfrak{g}$, $B^2(\mathfrak{g}, E) = \{ \omega : \mathfrak{g} \rightarrow E : \exists v \in E \text{ s.t. } \omega(x) = x \cdot v \}$.

Prop. $B^1(\mathfrak{g}, E) \subset Z^1(\mathfrak{g}, E)$.

Def. $H^1(\mathfrak{g}, E) = Z^1(\mathfrak{g}, E) / B^1(\mathfrak{g}, E)$.

Ext $\mathfrak{g}$ (k, E) = $H^1(\mathfrak{g}, E)$.

Ext $\mathfrak{g}$ (W, U) = $H^1(\mathfrak{g}, \text{Hom}_K(W, U))$

Ex. $H^1(\mathfrak{g}, \mathfrak{g}) = D^*(\mathfrak{g})$. $a \in B^1(\mathfrak{g}, \mathfrak{g})$. $\exists y \in \mathfrak{g}$ s.t. $a(x) = [x, y] = -\text{ad}_y(x)$.

Ex. if $\mathfrak{g}$ is abelian, $H^1(\mathfrak{g}, \mathfrak{g}) = 0$, $H^1(\mathfrak{g}, k) = 0$.

Prop. Chevalley complex $0 \rightarrow E \xrightarrow{d} \text{Hom}_K(\mathfrak{g}, E) \xrightarrow{d} \text{Hom}_K(\wedge^2 \mathfrak{g}, E) \xrightarrow{d} \dots$

$$df(x_1 \wedge \dots \wedge x_k) = \sum_{i=1}^k (-1)^{i-1} x_i \cdot f(x_1, \dots, \hat{x}_i, \dots, x_k) - \sum_{i < j} (-1)^{i+j} f([x_i, x_j], x_1, \dots, \hat{x}_i, \dots, \hat{x}_j, \dots, x_k)$$

Prop. $d_0 : B^1(\mathfrak{g}, E) \rightarrow B^2(\mathfrak{g}, E)$, $k_0 : B^1(\mathfrak{g}, E) \rightarrow B^2(\mathfrak{g}, E)$, $k_0 : B^1(\mathfrak{g}, E) \rightarrow B^2(\mathfrak{g}, E)$.

computes cohomology of $\mathfrak{g}$ $H^i(\mathfrak{g}, E)$ $\text{Ext}_{\mathfrak{g}}^i(k, E)$